

NAYEBARE MICHEAL

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<https://scholar.google.com/citations?user=xXIZhPAAAAAJ&hl=en>

RESEARCH INTERESTS

Human-Centered AI, Resilient AI Sociotechnical Systems, Community-Based Computing, and Generative Justice. My current research explores the design of AI systems that are fair, equitable, transparent, and community-driven. I integrate machine learning with ethical AI governance to build sociotechnical ecosystems that merge technical capability with social responsibility and inclusiveness.

EDUCATION

PhD Candidate, School of Information, **University of Michigan**. Dissertation research focuses on designing resilient AI systems for equitable, community-driven computational design and Generative Justice.

MS in Information Technology, **Carnegie Mellon University**, 2018.

BSc in Software Engineering and Business Administration, Uganda Christian University, Mukono, Uganda, 2012.

KEY PROJECTS & ACHIEVEMENTS

Ubuntu-AI Initiative, University of Michigan and Open AI. Co-developed a participatory bottom-up AI training and governance framework that integrates community participation and democratized control. Collaborated with OpenAI's Democratic Inputs team to pilot collective feedback loops in model training, enhancing stakeholder inclusion and transparency in LLMs. Featured in OpenAI's "Democratic Inputs to AI" grant program update, 2023.

<https://openai.com/index/democratic-inputs-to-ai-grant-program-update/>

IoT Tea Oxidation Modeling Research, Carnegie Mellon University. Designed and deployed a real-time IoT-ML agricultural analytics pipeline using Apache Kafka, processing over 1GB of sensor data to model and optimize tea oxidation processes. Delivered robust analytics pipelines that informed sustainable agriculture models for large-scale food production systems.

Industrial IoT Edge Gateways Development, fieldcloud SAS. Built Python/C-based embedded software CI/CD pipelines for industrial monitoring systems, improving deployment and firmware

over-the-air (FOTA) updates. Reduced deployment time by 70% and improved system monitoring by 30%, significantly enhancing scalability and remote maintainability across industrial IoT networks. These were significant transitions from the manual physical firmware development and deployment processes.

PROFESSIONAL EXPERIENCE

Back-End Infrastructure Engineer (AWS, Linux OS), Enables Platforms, Inc., New York, NY, 2022–2023. Architected and maintained scalable backend infrastructure using AWS Lambda, EC2, and S3. Developed Node.js SDK integrations and improved data processing efficiency through serverless architecture optimization.

Engineering Systems Integration Consultant – IoT Systems, fieldcloud SAS, Grenoble, France, 2020–2022. Integrated and customized embedded systems for industrial IoT gateways. Developed automated Linux kernel deployment pipelines and delivered converging IT and operational technology (OT) solutions.

Data & Software Engineer, Carnegie Mellon University Incubation Lab, Kigali, Rwanda, 2019–2022. Led the design of ETL and streaming pipelines with Apache Kafka for IoT-based agricultural analytics. Collaborated with interdisciplinary research teams to implement predictive models for optimizing oxidation processes in tea production in Nyabihu tea factory.

SELECTED PUBLICATIONS

Nayebare, M., Eglash, R., Kimanuka, U., Baguma, R., Mounsey, J., & Ciira, W. M. (2023). Interim report for Ubuntu-AI: A bottom-up approach to more democratic and equitable training and outcomes for machine learning. Conference: Democratic Inputs for AI.

Eglash, Ron, and others, 'Ubuntu AI: Machine Learning for Regenerative Design Ecologies' (21 May 2025), in Philipp Hacker (ed.), Oxford Intersections: AI in Society (Oxford, online edn, Oxford Academic, 20 Mar. 2025 -), <https://doi.org/10.1093/9780198945215.003.0138>, accessed 9 Nov. 2025

Eglash, R., Nayebare, M., Robinson, K., Robert, L., Bennett, A., Kimanuka, U., & Maina, C. (2024). AI governance through fractal scaling: Integrating universal human rights with emergent self-governance. *AI & Society*.

Porter Robinson, K., Eglash, R., Robert, L., Bennett, A., Guzdial, M., & Nayebare, M. (2025). Computing for Community-Based Economies: A Sociotechnical Ecosystem for Democratic, Egalitarian and Sustainable Futures. *arXiv:2504*.

Nayebare, M. (2019). Artificial intelligence policies in Africa over the next five years. *XRDS: Crossroads, The ACM Magazine for Students*, 26(2), 50–54. Kwizera, E., Mico, D., Nayebare, M.,

Garba, A. A., Saint, M., & Deen, L. G. (2018, March). The Impact of over the Top Service Providers in the Rwandan Telecommunications Market: An Analysis of Business Models. In International Conference on Innovations and Interdisciplinary Solutions for Underserved Areas (pp. 62-71). Cham: Springer International Publishing.

MENTORSHIP

ARM Ambassador Program, ARM UK. Delivered MicroPython and RP-Nano2040 training to over 230 engineering students in 10+ African countries, emphasizing low-cost, promoting sustainable ML education in resource-constrained environments. The articles have received over 530 reads. Given that I was volunteering, I tried to create as much documentation as I could. See my substack: <https://michealnayebare.substack.com/p/getting-started-with-arduino-nano>.